Group 6

Semantic Technologies

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BADS: Polimi GSoM

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Project Roadmap

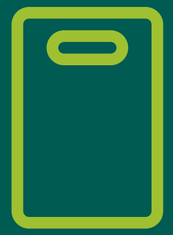
Domain

Semantic Technologies course, part of the BADS Program at GSoM



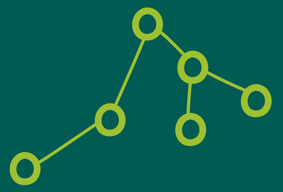
Annotation

Schema.org markup in the course HTML page



Graph

Extracted turtle graph (.ttl) from RDFa through *EasyRdf*



Queries

Queries with SPARQL covering *SELECT, FILTER ETC.*



Schema - Course

2 credits | code ST-BADS-GSOM-POLIMI | language: en

Polimi GSoM
Provider

4.6/5 - 134 Reviews
aggregateRating

HasCourseInstance

Instance 2026
13-19 May - Milan - Onsite
15hr Workload
instructor → Matteo Belcao

Instance 2027
1 May 2027
(sparse - startDate only)

Offers

€0
free tier

€800
paid tier

Prerequisites

- basic programming
- basic logic
- basic databases

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HTML With RDFa Annotation

RDFa Attributes

- **Vocab**: sets default namespace
- **Typeof**: declares a new entity's type
- **Resource**: names entities (URI fragment)
- **Property**: attaches a value to the subject
- **Content**: overrides the visible text
- **Datatype**: types literals (xsd:date, …)

```
<body vocab="schema.org" typeof="Course">
  <h1 property="name">Semantic Technologies</h1>
  <span property="hasCourseInstance" typeof="CourseInstance"
        resource="#instance2026">…</span>
  <span property="numberOfCredits"
datatype="xsd:integer">2</span>
  <span property="name" content="Politecnico di Milano">
        Polimi GSoM</span>
```



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HTML

```html
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Semantic Technologies | Group 6 | Cambier - Cobuccio, Ferrara - Tissino</title>
</head>
<body vocab="https://schema.org/" typeof="Course">

  <h1 property="name" content="Semantic Technologies">Semantic Technologies | Group 6 | Cambier - Cobuccio - Ferrara - Tissino</h1>
  <h2>Course Description</h2>

  <p property="description">This course is part of the Master BADS taught at
     <span property="provider" typeof="CollegeOrUniversity">
     <span property="name" content="Politecnico di Milano - Graduate School of Management">Polimi GSoM</span>,
     the Politecnico di Milano Business School</span>. <br>
     Students will learn about the logic behind RDF and OWL, learning the use of
     SPARQL and how to apply these concepts by using Ontop in the field of big data.</p>

  <h2>General Information</h2>

  <p><span property="hasCourseInstance" typeof="CourseInstance" resource="#instance2026">
     This course is delivered in <span property="location">Milan, Italy</span>,
     <span property="courseMode" content="Onsite">in-person</span> only.
     However, students may watch the recordings that will be made available online afterwards.<br>
     The professor is <span property="instructor" typeof="Person"><span property="name">Matteo Belcao</span>.
     He can be contacted at <span property="email">matteo.belcao@quantiaconsulting.com</span> throughout the course.</span></span><br>
     The whole course is taught in <span property="inLanguage" content="en">English</span>.
     The number of credits for this course is <span property="numberOfCredits" datatype="xsd:integer">2</span>.</p>

  <h2>Course Structure</h2>

  <p>This course is composed of three classes:
    "Introduction to the 'variety dimension' of Big Data - The RDF data model",
    "Publishing smart data using RDF", and
    "OWL ontological language - Ontology-based data integration",
    <span about="#instance2026">for a total of <span property="courseWorkload" content="PT15H">approximately 15 hours</span>.
    The first class is taught on <span property="startDate" content="2026-05-13" datatype="xsd:date">13 May 2026</span>,
    the second on 18 May 2026, and the last on <span property="endDate" content="2026-05-19" datatype="xsd:date">19 May 2026</span>.</span>
    The exam will take place on 28 May 2026, starting at 1 PM.</p>

```

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HTML

```
41      <p>The course material consists of six slideshows, each corresponding to a specific topic:</p>
42
43      <ul>
44        <li>Smart Data and Smart Machines</li>
45        <li>RDF Data Model</li>
46        <li>Querying the Semantic Layer with SPARQL</li>
47        <li>RDFa Syntax for the Web</li>
48        <li>Conceptual Modeling with RDFS and OWL</li>
49        <li>Ontology-Based Data Access Using Ontop</li>
50      </ul>
51
52      <h2>Administration</h2>
53
54      <p>This course is part of the Politecnico di Milano course catalog. The course code (remember it!) is:
55        <strong property="courseCode">ST-BADS-GSOM-POLIMI</strong>. <br>
56         The course is <span property="offers" typeof="Offer">included in the BADS Master tuition (€0) (<span property="priceCurrency" content="EUR">€</span>
57         <span property="price" datatype="xsd:decimal">0</span>)</span>; <span property="offers" typeof="Offer">for external participants the fee is
58         <span property="priceCurrency" content="EUR">€</span><span property="price" datatype="xsd:decimal">800</span></span>. <br>
59         <span property="hasCourseInstance" typeof="CourseInstance" resource="#instance2027">
60         The next edition of the course is planned for <span property="startDate" content="2027-05-01" datatype="xsd:date">May 2027</span> with updated content and teaching methods.
61
62      <h2>Prerequisites</h2>
63
64      <p>In order to participate and get the most out of this course, students should have a basic knowledge of the following topics:
65        <span property="coursePrerequisites">basic programming</span>,
66        <span property="coursePrerequisites">basic logic</span>, and <span property="coursePrerequisites">basic databases</span>.</p>
67
68      <h2>Reviews and Reputation</h2>
69
70      <p>This course is part of the BADS Master's program, which is among the top 50 Master's in the world. More specifically, this course
71        has an overall rating of <span property="aggregateRating" typeof="AggregateRating"><span property="ratingValue" datatype="xsd:decimal">4.6</span> out of
72        <span property="bestRating" datatype="xsd:decimal">5</span> based on more than <span property="ratingCount" datatype="xsd:integer">134</span> reviews</span>.</p>
73
74    </body>
75    </html>
```

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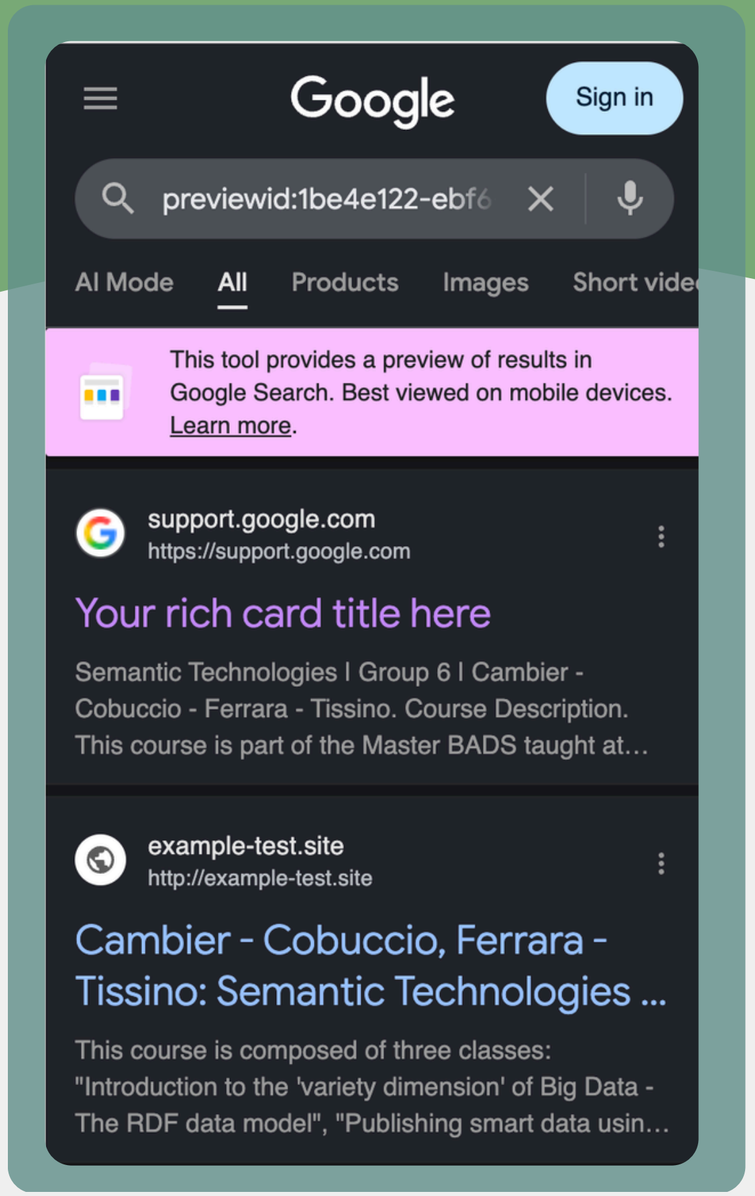
RDFa

```
@prefix rdfa: <http://www.w3.org/ns/rdfa#> .
@prefix ns0: <https://schema.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

<http://njh.me/> rdfa:usesVocabulary <https://schema.org/> .
<http://njh.me/#instance2026>
  a <https://schema.org/CourseInstance> ;
  ns0:location "Milan, Italy" ;
  ns0:courseMode "Onsite" ;
  ns0:instructor [
    a ns0:Person ;
    ns0:name "Matteo Belcao" ;
    ns0:email "matteo.belcao@quantiaconsulting.com"
  ] ;
  ns0:courseWorkload "PT15H" ;
  ns0:startDate "2026-05-13"^^xsd:date ;
  ns0:endDate "2026-05-19"^^xsd:date .

<http://njh.me/#instance2027>
  a ns0:CourseInstance ;
  ns0:startDate "2027-05-01"^^xsd:date .
```

Rich Result Test



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Select Query

```
SELECT ?name ?credits ?code ?language
WHERE {
  ?course a schema:Course ;
          schema:name                ?name ;
          schema:numberOfCredits     ?credits ;
          schema:courseCode          ?code ;
          schema:inLanguage          ?language .
}
```

name	credits	code	language
"Semantic Technologies"	2	"ST-BADS-GSOM-POLIMI"	"en"

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Join Query

```
SELECT ?courseName ?instructorName ?instructorEmail
WHERE {
  ?course a schema:Course ;
          schema:name               ?courseName ;
          schema:hasCourseInstance ?instance .

  ?instance schema:instructor ?instructor .

  ?instructor schema:name  ?instructorName ;
              schema:email ?instructorEmail .

  FILTER(STR(?courseName) = "Semantic Technologies")
}
```

courseName	instructorName	instructorEmail
"Semantic Technologies"	"Matteo Belcao"	"matteo.belcao@quantiaconsulting.com"

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Filter values Query

```
SELECT ?courseName ?price ?currency
WHERE {
  ?course a schema:Course ;
          schema:name   ?courseName ;
          schema:offers ?offer .

  ?offer schema:price         ?price ;
         schema:priceCurrency ?currency .

  FILTER(?price > 0)
}
```

courseName	price	currency
"Semantic Technologies"	800.0	"EUR"

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Dates Filter Query

```
SELECT
    (STR(?instance) AS ?cleanInstance)
    (STR(?startDate) AS ?cleanStartDate)
WHERE {
  ?course a schema:Course ;
          schema:hasCourseInstance ?instance .

  ?instance schema:startDate ?startDate .

  FILTER(?startDate > "2026-12-31"^^xsd:date)
}
```

cleanInstance	cleanStartDate
"http://njh.me/#instance2027"	"2027-05-01"

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Optional Query

```
SELECT
  (STR(?instance) AS ?cleanInstance)
  (STR(?startDate) AS ?cleanStartDate)
  (STR(?endDate) AS ?cleanEndDate)
  (STR(?location) AS ?cleanLocation)
  (STR(?courseMode) AS ?cleanMode)
WHERE {
  ?course a schema:Course ;
          schema:hasCourseInstance ?instance .

  ?instance schema:startDate ?startDate .

  OPTIONAL { ?instance schema:endDate    ?endDate . }
  OPTIONAL { ?instance schema:location   ?location . }
  OPTIONAL { ?instance schema:courseMode ?courseMode . }
}
```

cleanInstance	cleanStartDate	cleanEndDate	cleanLocation	cleanMode
"http://njh.me/#instance2026"	"2026-05-13"	"2026-05-19"	"Milan, Italy"	"Onsite"
"http://njh.me/#instance2027"	"2027-05-01"			

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Aggregate Query

```
SELECT
  ?courseName
  (COUNT(?offer) AS ?numOffers)
  (AVG(?price)   AS ?avgPrice)
  (MIN(?price)   AS ?minPrice)
  (MAX(?price)   AS ?maxPrice)
WHERE {
  ?course a schema:Course ;
          schema:name   ?courseName ;
          schema:offers ?offer .
  ?offer schema:price ?price .
}
GROUP BY ?courseName
```

courseName	numOffers	avgPrice	minPrice	maxPrice
"Semantic Technologies"	2	400.0	0.0	800.0

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Union Query

```
SELECT ?value
WHERE {
  { ?course schema:courseCode ?value }
  UNION
  { ?course schema:inLanguage ?value }
}
```

```
1 ------------------------
2 | value                |
3 ========================
4 | "ST-BADS-GSOM-POLIMI" |
5 | "en"                 |
6 ------------------------
7
```

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Limit Query

```
SELECT ?price
WHERE {
  ?course schema:offers ?offer .
  ?offer schema:price ?price .
}
ORDER BY ASC(?price)
LIMIT 1
```

```
---------
| price |
=========
| 0.0   |
---------

```

Thank You!